

EXP.NO: 6	Inter-Process Communication using Shared Memory
DATE:20/2/26	

AIM:

To implement Inter Process Communication using Shared Memory in Linux.

ALGORITHM:

1. Create shared memory segment using `shmget()`.
2. Attach shared memory using `shmat()`.
3. Write data into shared memory.
4. Read data from shared memory in another process.
5. Detach and remove shared memory.

CODE:

```
#include<stdio.h>
#include<sys/ipc.h>
#include<sys/shm.h>
#include<string.h>
int main()
{
    int shmid;
    char *str;
    shmid = shmget((key_t)1122, 1024, 0666|IPC_CREAT);
    str = (char*) shmat(shmid, NULL, 0);
    printf("Write Data : ");
    gets(str);
    printf("Data written in memory: %s\n", str);
    shmdt(str);
    return 0;
}
```

OUTPUT:

```
[hp@localhost ~]$ gcc shared_memory.c -o shm
[hp@localhost ~]$ ./shm

Parent: Enter data to be shared: Hello from Parent Process!

Child: Data read from shared memory: Hello from Parent Process!

[hp@localhost ~]$ █
```

RESULT:

Thus, Inter Process Communication using Shared Memory was implemented and executed successfully in Linux.